

TractorBeam: Personalized AI Sensemaking Support via Collaborative Machine Annotation

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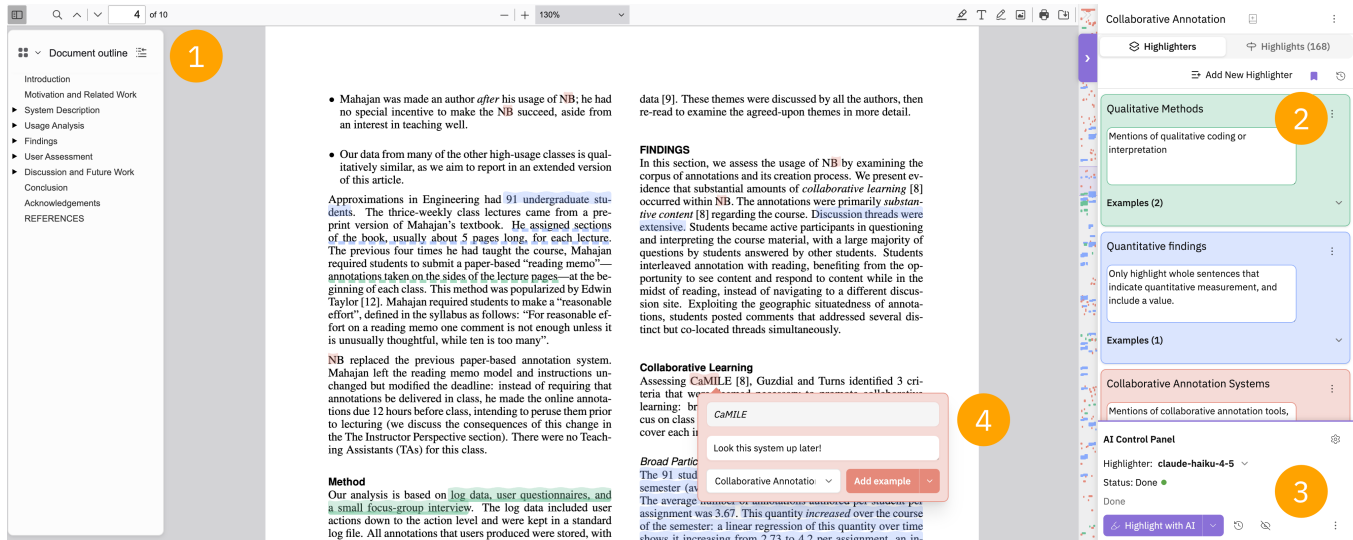


Figure 1: A screenshot of TractorBeam open on a paper, showing 1. The PDF viewer, where both human and LM annotations are displayed directly on the PDF; 2. The "highlighters" view, where users can specify their own schema for annotation; 3. The AI control panel, where users can configure and trigger AI highlighting; 4. An example modal dialog for users to accept or reject LM highlights before iteratively re-annotating. In the viewer, solid and dashed underlines indicate user-provided positive and negative examples, and squiggly highlights indicate AI highlights.

Abstract

Language model-based systems which allow asking questions of documents have become popular tools for sensemaking. Despite their implied capability, these systems still suffer from issues of factuality and provenance, while encouraging confirmatory, rather than exploratory, research. We present TractorBeam, a browser extension-based mixed-initiative system that uses collaborative annotation as an interface metaphor for sensemaking, re-framing language model (LM) outputs as suggested highlights in a process that we call *collaborative machine annotation*. This metaphor allows us to present LM results in-context on PDF documents, directly addressing concerns of provenance and factuality, while allowing users to iteratively construct mental schemas and queries for language models directly in the context of a document. In a preliminary

user study, all of our participants felt that TractorBeam enabled them to evaluate and iteratively improve the model's reflection of their intended highlighting, and several found suggestions that made them reconsider their original schema. TractorBeam suggests that systems that facilitate exploratory research on individual documents may lead to verifiable sensemaking for users and complement tools that work across broader corpora.

CCS Concepts

• **Human-centered computing** → **Collaborative interaction**; *Web-based interaction*; • **Computing methodologies** → **Information extraction**.

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1 Introduction

Large language models (LLMs) are increasingly being pitched as solutions that expand the scope of sensemaking, i.e. organizing and constructing meaning from information [14, 20]. Systems like ScholarQA [2] and DocWrangler [23] facilitate complex processing over collections of documents, and LM-based assistance for qualitative coding is an active area of research [9, 15, 22, inter alia]. Though these systems allow users to specify complex and specific intents, they nonetheless raise a number of concerns. First, where the output takes the form of generated text, verifying the provenance and accuracy of the information in the generated text remains a challenge. While citations have often been presented as a fix for this problem, Ding et al. [7] found that the presence of citations can increase trust in results even as following those citations decreases it; Onweller et al. [17] find that even where citations are referentially correct, the cited sources often do not support the evidence presented. Earlier sensemaking systems like Scim [8] or PaperPlain [3], though limited by the technology of the time and therefore being statically defined, do not suffer the provenance issue by displaying their suggestions as in-context highlights, which makes evaluating accuracy and provenance much simpler. More generally, LM-based sensemaking tools typically require users to know up-front what they are looking for in the form of e.g. initial question to a system like ScholarQA. However, as is well-documented in the sensemaking literature, questions of this type are often iteratively constructed and refined in the process of engaging with documents [4, 12, 18, 21]. While LM-based tools can support exploring already well-formed research questions, we argue that more work is needed for LM-based tools to better support open-ended exploratory, rather than confirmatory, work.

To better support LM-assisted, iterative exploration, we present TractorBeam, a browser extension-based system that conceptualizes sensemaking assistance as a process of collaborative annotation on document content between a user and an LM, a process that we call *collaborative machine annotation*. TractorBeam draws on the extensive study of collaborative annotation in educational settings [5, 6, 11, 13, 16, 24, 26] that establish that annotation, both collaborative and individual, can increase engagement in reading. The placement of annotations directly on the originating documents also allows users to continue their iterative and interpretive reading practices, while at the same time offering them the ability to engage with annotations as suggested points of interest while interactively evaluating and editing them, as in systems like PaTAT [10], ChainForge [1], or Raggy[19]. As users develop deep familiarity with the models they build, these models can also be extended to support corpus-level annotation as in systems like DocWrangler[23].

2 System Design

Figure 1 shows the view that a user has when they launch TractorBeam: a PDF viewer that also displays highlights, and a sidebar where users can specify annotation intent, navigate highlights, and configure and start AI highlighting. User annotation in TractorBeam is schema-driven: highlights are assigned to categories, which we call highlighters. To reflect the iterative and cyclical nature of sensemaking processes [18], highlighters can be added, deleted, or edited at any point, and highlights can be reassigned between

highlighters. TractorBeam also uses an event sourcing model for data storage, meaning that schemas are versioned by default, allowing users to experiment with schemas and roll back to a previous version if necessary.

LM-based annotation in TractorBeam is derived automatically from the user's schema. This architecture frees users from thinking about best practices for prompting language models, which can be a challenge [25], and enables future work like offline prompt optimization based on user annotations. All runs of the LM-based highlighters are also stored, and users can see their run history, and apply those runs back to the PDF at any time. Runs are stored linked to schema versions, and both are given friendly names, allowing users to reason about what schema changes might contribute to particularly performant LM highlighting runs. Users can additionally choose only to run LM-based highlighting on subsets of the document, enabling rapid iteration on highlighters.

Highlights themselves are visualized on the PDF to communicate their status: user annotations are underlines, with solid and dashed lines communicating positive and negative examples, respectively; LM-based annotations are communicated as squiggles to communicate uncertainty, and to draw users' eyes to potentially new parts of the document. Users can indicate that model highlights should be considered positive or negative examples for a highlighter directly from the PDF viewer. To allow users to use the highlights to guide their reading, especially in long documents, we present both a "Highlights" view, which allows for filtering the highlights by metadata, and a minimap that overlays the scroll bar, which visualizes global highlight position on the document at a glance. We intend for users to see this as a sort of semantic index, allowing them to discover structure relevant to their interests in documents that may not correspond with the authors' view. As users develop their models, they can also view annotations across documents in the "manager" view, and export data for their own sensemaking, or to train more efficient models to scale out their annotation.

3 Preliminary User Study

We conduct a preliminary user study with 6 participants across different fields: environmental policy scholarship (P0, P5), materials science (P1, P4), materials informatics (P2), broadcast journalism (P3). Our participants were all adults with over 3 years of experience in their fields (mean: 16.1; median: 10.75), who read and analyzed PDFs as part of their regular work. Participants were shown a demonstration of the tool before a think aloud session in which they used TractorBeam to read PDF documents representative of their own work. Our study was IRB-approved as STUDY2026_00000272.

Participants' existing reading processes varied—no two participants shared an identical schema, even when from the same domain. Many participants found the tool and its highlights useful in their reading, both confirmatory and exploratory. P3 noted how the AI highlights would help read documents under time constraints, and P0 discussed how the highlights helped "*parse out the core elements*" of a document. All participants felt that they could improve the LM predictions over time, as in the case of P1 saying "*just needed one negative example of the first author, and it corrected that on another attempt*". While the initial LM results did not always prove

useful, participants could successfully develop models that worked through iterating using TractorBeam’s provided affordances.

4 Future Work

We view TractorBeam as a preliminary tool aimed at scaling up users’ detailed, interpretive and idiosyncratic reading processes, without losing those qualities at scale. We envision two directions of future work: capturing further detail in users’ handling of documents, including multimodal information like tables, and generalizing out, such that users can work with entire document collections after having developed their schemas in TractorBeam.

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